

Deepan Kulandaisami

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Profile

Software Engineer with **1.5+ years of production experience** building **LLM applications, RAG systems, AI agents, and Python backend systems**. Hands-on experience with **FastAPI, Django, LangGraph, LangChain, CrewAI, PostgreSQL, vector databases, document processing, tool calling, and LLM integrations**. Experience across the development lifecycle including **AI implementation, backend development, testing, debugging, deployment, and production support**.

Education

National Institute of Technology Karnataka (NITK)

B.Tech in Electrical & Electronics Engineering; Minor in Mathematical & Computational Sciences

Mangalore, India

2020 – 2024

Professional Experience

Associate Software Developer – AI Full Stack

Apr. 2025 – Sep. 2026

BONbLOC AI

- Developed and maintained production software across **AI implementation, backend development, frontend integration, testing, deployment, debugging, and production support** in a startup environment with end-to-end ownership of assigned systems.
- Designed and implemented Python backend services using **FastAPI, and Django**, developing APIs, business workflows, database integrations, authentication, external-system integrations, and AI service integrations.
- Built and orchestrated AI workflows using **LangGraph, LangChain, and CrewAI**, integrating LLMs, tools, databases, external APIs, retrieval systems, and stateful agent workflows.
- Implemented document and knowledge-processing pipelines involving **PDF extraction, OCR, document chunking, embeddings, vector search, and LLM-based retrieval workflows**.
- Developed database-driven AI workflows using **PostgreSQL**, including schema handling, SQL generation and validation, query execution, error handling, and retry mechanisms.
- Implemented real-time and asynchronous backend communication using **WebSockets, Server-Sent Events (SSE), and Celery** for streaming responses, background processing, and long-running tasks.
- Implemented authentication and authorization using **Keycloak and Azure AD/MSAL**, integrating identity and access control with backend applications.
- Containerized applications using **Docker** and worked on production deployments using **Azure Container Apps and Azure Database for PostgreSQL – Flexible Server**.
- Performed **testing, debugging, failure analysis, root-cause analysis, API validation, and production issue resolution**, using tools including Postman and application logs.
- Contributed to **requirements analysis, technical design, estimation, implementation, code reviews, documentation, and Agile Scrum delivery**, taking ownership across the software development lifecycle.

Projects

Internal HR Policy Chatbot – End-to-End RAG | Django, LangChain, Vector Databases, LLMs

- **Problem:** Enable employees to query internal company policy documents using natural language while handling PDFs containing structured, unstructured, and scanned content.
- **Approach:** Implemented an end-to-end **RAG pipeline** covering document ingestion, PDF extraction, OCR, preprocessing, chunking, embeddings, retrieval, and LLM-based response generation. Used **PyMuPDF, Tesseract, and MonkeyOCR** for document processing and evaluated multiple vector stores including **FAISS, Milvus, and Qdrant**.
- **Solution:** Integrated retrieval and generation into the **Django** chatbot backend and worked on retrieval optimization, RAG evaluation, and scaling to improve the reliability of document-grounded responses.

CarryEx Supply-Chain SQL Agent | FastAPI, LangGraph, PostgreSQL, LLMs

- **Problem:** Enable natural-language querying over a large supply-chain PostgreSQL database containing numerous tables and complex relationships, without requiring users to write SQL.
- **Approach:** Built a **FastAPI and LangGraph** agent that discovers database schema information, retrieves relevant schema context, generates SQL using an LLM, validates the query, and executes it against PostgreSQL.
- **Solution:** Implemented **error-driven SQL correction and retry workflows**, feeding database errors back into the agent for query correction. Added flexible entity/name searches and stateful orchestration across schema discovery, generation, execution, and recovery.

Proposal Generation Agent | FastAPI, LangGraph, LLMs, Document Processing

- **Problem:** Automate the creation of client proposals from requirements gathered through conversation or information contained in client-provided documents.
- **Approach:** Developed a **FastAPI-based agentic workflow** for requirements extraction, information processing, content generation, and proposal structuring using LLMs.
- **Solution:** Generated structured proposals and implemented export workflows supporting **PDF, PPT, and DOC** formats, reducing manual effort involved in preparing client proposals.

Onelign Edge Agentic Chatbot | *FastAPI, LangGraph, React.js, Tool Calling*

- **Problem:** Provide a conversational interface for interacting with the existing Onelign Edge platform through natural-language requests.
- **Approach:** Built a **FastAPI chatbot backend** with agentic tool calling using **LangGraph**, exposing existing **Spring Boot REST endpoints as tools** available to the agent.
- **Solution:** Implemented the **React.js UI**, conversational workflow, tool selection, API execution, and response orchestration, allowing users to perform platform operations through natural language.

Cricket Analytics Application – CSK | *FastAPI, Machine Learning, Data Analytics*

- **Problem:** Derive meaningful performance insights from cricket data to support analytical understanding of players, teams, and match performance.
- **Approach:** Applied **machine-learning and data-analysis techniques** to cricket datasets, developing analytical workflows for feature processing, modelling, and insight generation.
- **Solution:** Built a **FastAPI-based application** exposing the resulting analytics and insights through backend services.

Conversational Voice Processing | *FFmpeg, OpenAI Whisper, Sarvam AI*

- **Problem:** Extend the internal HR chatbot with voice-based interaction while integrating different speech-processing services into the conversational workflow.
- **Approach:** Implemented audio processing using **FFmpeg** and integrated **OpenAI Whisper** for speech-to-text and **Sarvam AI** for voice capabilities.
- **Solution:** Connected audio processing and speech services with the chatbot backend to support voice-driven conversational interactions.

Technical Skills

AI / GenAI: LLM Applications, AI Agents, LangGraph, LangChain, CrewAI, Tool Calling, OpenAI, Hugging Face, Ollama
RAG & Retrieval: End-to-End RAG Pipelines, RAG Evaluation, Retrieval Optimization, Embeddings, Vector Search, FAISS, Milvus, Qdrant
LLM Engineering: LLM Evaluation, Prompt Evaluation, Agent Workflows, Structured Outputs, Error Handling, Retry Strategies
Python Backend Systems: Python, FastAPI, Django, Django REST Framework, Celery, REST APIs, WebSockets, Server-Sent Events
Data & Document Processing: PostgreSQL, SQL, Schema Design, SQL Generation, Query Validation, PyMuPDF, Tesseract, MonkeyOCR
Voice & Media: FFmpeg, OpenAI Whisper, Sarvam AI, Audio Processing, Speech-to-Text
Frontend: JavaScript, TypeScript, React.js, HTML, CSS
Cloud & Systems: Docker, Azure, Azure Container Apps, Azure Database for PostgreSQL, Linux, Git, Distributed Computing, Parallel Computing
Security: Keycloak, Azure AD, MSAL, Authentication, Authorization, Access Control
C++: C++, OOP, STL, RAII, Smart Pointers, Multithreading, Concurrency, Mutexes, Memory Management, Data Structures & Algorithms
Java: Java, Spring Boot, Apache Kafka
Engineering: Requirements Analysis, Technical Design, Debugging, Testing, Code Reviews, Root-Cause Analysis, Agile/Scrum, SDLC

Certifications

Microsoft Certified: Azure Developer Associate (AZ-204) – Certification ID: **E83442-7985BF**; Expires: **30 Sep. 2027**
Microsoft Certified: Azure Fundamentals (AZ-900) – Certification ID: **3672EB-85BFR0**; **Does not expire**